

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Numerical

A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

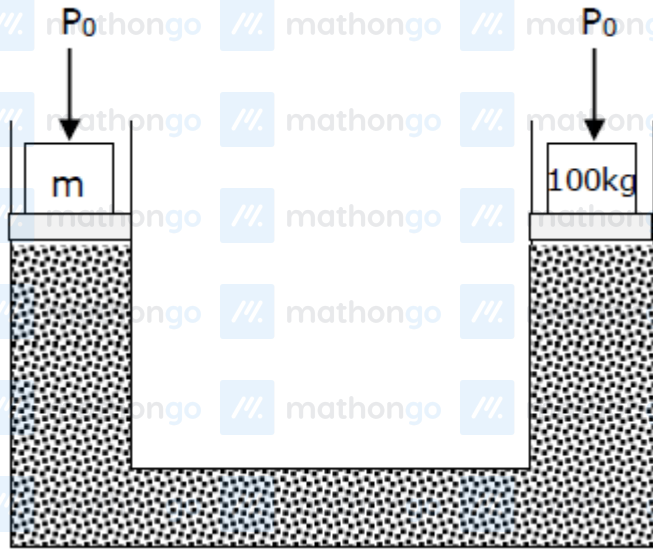
Q2: 26 Feb (Shift 1) - Single Correct

A large number of water drops, each of radius r , combine to have a drop of radius R . If the surface tension is T and mechanical equivalent of heat is J , the rise in heat energy per unit volume will be :

- (1) $\frac{2T}{rJ}$
- (2) $\frac{3T}{rJ}$
- (3) $\frac{2T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$
- (4) $\frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$

Answer Key**Q1 (25600)****Q2 (4)**

Q1 (25600)



Atmospheric pressure P_0 will be acting on both the limbs of hydraulic lift.

Applying pascal's law for same liquid level

$$\Rightarrow P_0 + \frac{mg}{A_1} = P_0 + \frac{(100)g}{A_2}$$

$$\Rightarrow \frac{Mg}{A_1} = \frac{(100)g}{A_2} \Rightarrow \frac{m}{100} = \frac{A_1}{A_2} \dots (1)$$

Diameter of piston on side of 100 kg is increased by 4 times so new area $= 16 A_2$

Diameter of piston on side of (m)kg is decreasing

$$A_1 = \frac{A_2}{16}$$

(In order to increasing weight lifting capacity, diameter of smaller piston must be reduced)

$$\text{Again, } \frac{mg}{\left(\frac{A_1}{16}\right)} = \frac{M'g}{16A_2} \Rightarrow \frac{256m}{M'} = \frac{A_1}{A_2}$$

$$\text{From equation (1)} = \frac{256m}{M'} = \frac{m}{100} \Rightarrow \therefore M' = 25600 \text{ kg}$$

Q2 (4)

R is the radius of bigger drop. is the radius of n water drops. Water drops are combined to make bigger drop.

So,

Volume of n drops = volume of bigger drop

$$n \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3$$

$$R = rn^{1/3} \Rightarrow n = \left(\frac{R}{r} \right)^3$$

$$\Delta U = T (\text{Change in surface area}) \Delta U = T (n4\pi r^2 - 4\pi R^2)$$

$$\Delta U = 4\pi T \left[\left(\frac{R}{r} \right)^3 r^2 - R^2 \right] \Rightarrow \frac{4\pi T \left(\frac{R^3}{r} - R^2 \right)}{J}$$

$$\frac{\Delta U}{V} = \frac{4\pi T \left(\frac{R^3}{r} - R^2 \right)}{J \times \frac{4}{3} \pi R^3} = \frac{3T}{J} \left[\frac{1}{r} - \frac{1}{R} \right]$$